
CF16: ELECTROWEAK FIBER SPLITTING, CHIRAL WEAK COUPLING, NEUTRAL GAUGE MIXING, AND BROKEN GAUGE–SCALAR STRUCTURE IN VDM

COMPLETE FORMALISM

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Abstract

This paper derives the minimal electroweak realization of the VDM Complete Formalism as a descendant of the Primitive Bifurcation Law rather than as an imported Standard Model sector. The derivation uses only already-earned canon: carrier and local phase redundancy (CF 00), tachyonic vacuum/interface forcing (CF 03), chiral boundary localization (CF 08), multi-mode gauge hosting from fiber structure (CF 09), chirality/orientation (CF 13), stationary articulation-cost selection (CF 14), and zero-cost conservation and symmetry breaking (CF 15). From these inherited structures the paper derives the minimal realized electroweak-hosting fiber, the restriction of the non-abelian weak sector to the active chiral boundary family, the persistence of an abelian neutral companion burden, neutral-sector mixing into one massless and one massive direction, the charged/neutral mass relation, and the physical scalar as the normal mode of the broken electroweak interface.

The central claim is structural: once the already-earned abelian phase redundancy, the minimal two-mode internal rotation sector, and the oriented boundary-hosted chiral matter sector are realized together, the electroweak pattern is forced. Standard electroweak language appears only as a downstream recognition layer: the left-acting two-mode non-abelian sector is identified with the weak sector, the abelian neutral companion with the hypercharge sector, the massless neutral residual direction with electromagnetism, the neutral diagonalizing angle with the Weinberg angle, the charged/neutral mass law with the tree-level W/Z relation, and the normal interface mode with the Higgs scalar. The electroweak architecture is closed here as a completed structural/tree-level derivation; later papers may refine already-derived relations, but they do not own, reopen, or repossess them.

Keywords electroweak unification, chiral weak coupling, neutral mixing, residual electromagnetism, Higgs interface mode, primitive bifurcation, domain-wall fermions, gauge hosting

1 Introduction

Motivation

CF 16 is the next realized gauge/matter closure after CF 15. Once CF 15 closes zero-cost directions, conserved burdens, symmetry breaking, residual unbroken directions, and the tangential/normal split of a broken sector, the next forced question is concrete: what broken gauge/matter pattern appears when the

already-earned chiral boundary-hosted matter sector and the already-earned multi-mode gauge-hosting fiber are written at the same structural level?

This paper answers that question for the electroweak layer. It does not import the Standard Model as a primitive object. Instead it asks what realized descendant is forced when the following previously earned facts are jointly admitted: a complex carrier with local phase redundancy, a minimal two-mode internal rotation sector, a chiral/oriented boundary-hosted active matter sector, and a tachyonically forced vacuum/interface order parameter. The answer is the minimal electroweak-hosting fiber and its broken/unbroken decomposition.

What This Work Closes

This paper closes the following structural burdens:

- **Minimal electroweak hosting** as the smallest realized matter/gauge fiber that can simultaneously bear the inherited abelian phase direction, a minimal non-abelian internal rotation, a chiral active boundary family, and a tachyonically forced order parameter.
- **Chiral weak coupling** as a consequence of domain-wall localization and oriented chirality rather than an imported representation assignment.
- **Neutral mixing, Weinberg-angle closure, and residual electromagnetism** as the forced diagonalization of the broken neutral sector into one massless residual direction and one massive neutral direction, with the unique diagonalizing angle θ_{EW} closed as the structural/tree-level electroweak mixing angle of the minimal realized layer.
- **Broken gauge–scalar structure and W/Z closure** as the charged/neutral mass relation, including the structural tree-level law $m_W = m_Z \cos \theta_{EW}$, and the survival of only the normal electroweak interface mode as a physical scalar.

What This Work Is Not

- This is *not* a flavor paper, a confinement paper, or a radiative corrections paper.
- This is *not* a precision radiative-corrections paper; those descendants belong to later closures once the present structural electroweak layer is fixed.
- This is *not* an imported Standard Model derivation; standard electroweak language appears only as downstream recognition of already-earned VDM structure.

2 Background: Canon Sources and the Electroweak Hosting Problem

Statement of the Problem

Definition 1 (Electroweak hosting problem). The electroweak hosting problem is to determine the minimal realized fiber, charge structure, and broken/unbroken sector decomposition forced when the following canon objects are admitted simultaneously: the complex carrier with local abelian phase redundancy, a two-mode internal rotation sector, a chiral boundary-hosted active matter family, and a tachyonically forced interface order parameter.

The question is not whether one can label sectors by the later language of $SU(2)_L$, hypercharge, photon, Z , or Higgs. The question is whether those labels are the inevitable coarse recognition of a uniquely forced VDM hosting structure.

Canon Sources and What They Provide

Every proof in this paper starts from canon already earned in the CF stack. Standard electroweak language is never a proof step.

1. **A(-1) and CF000.** These license primitive non-discharge, unresolved two-pole opposition, same-domain saturation forcing orthogonal re-articulation, and the contradiction of invariant non-bearing.
2. **CF00.** This licenses the carrier domain, smooth variation structure, complex fiber, local abelian phase redundancy, and quotient-physical representative logic.

3. **CF03.** This licenses tachyonic instability of a symmetric hosting configuration and the forced relocation of burden onto a vacuum/interface structure when symmetric hosting can no longer stably bear the invariant.
4. **CF08 and CF13.** These license boundary-localized chiral modes, asymptotic separation of the opposite-handed partner, and chirality as a real orientation asymmetry rather than a removable bookkeeping choice.
5. **CF09.** This licenses gauge hosting from fiber structure, the abelian phase direction, and the availability of a multi-mode internal rotation sector once more than one degenerate fiber mode is admitted.
6. **CF14 and CF15.** These license stationary articulation-cost selection, zero-cost directions, residual unbroken directions after symmetry breaking, and the tangential/normal split of broken sectors.

Inherited Structural Facts

The following facts are used throughout the paper.

1. The primitive invariant bears no location mark, no axis mark, and no preferred handedness before later realized structure is admitted.
2. The carrier already supports a local abelian phase redundancy (CF 00), so every downstream gauge-hosting realization inherits an abelian neutral companion direction.
3. A symmetric bulk hosting configuration may fail under tachyonic instability and relocate burden onto an interface or wall (CF 03, CF 08).
4. The active matter modes realized on that interface are chiral and oriented; the opposite-handed partner is not hosted in the same active boundary family (CF 08, CF 13).
5. Once more than one degenerate fiber mode is admitted, internal rotation is not exhausted by the abelian phase direction; a higher multi-mode internal rotation sector is available (CF 09).
6. A broken sector splits into tangential directions, which are gauge-absorbed or residual zero-cost directions depending on the sector, and a normal direction, which remains as the physical scalar interface mode (CF 15).

3 Formal Concepts

Core Quantity A: Active Chiral Matter Sector

Definition 2 (Active chiral boundary sector). The *active chiral boundary sector* is the realized family of boundary-hosted fermionic modes localized by the wall/interface construction of CF 08 and selected by the oriented handedness structure of CF 13. It is the matter sector that actually inhabits the weak-hosting boundary layer.

Definition 3 (Weak doublet-hosting pair). A *weak doublet-hosting pair* is the minimal two-mode family inside the active chiral boundary sector on which a single non-abelian internal rotation acts transitively while preserving the common hosting norm.

Core Quantity B: Neutral Companion and Electroweak Order Parameter

Definition 4 (Neutral companion burden / hypercharge sector). The *neutral companion burden* is the abelian fiber burden inherited from the already-earned local phase redundancy of CF 00. At the electroweak layer it is the unique one-dimensional neutral companion sector that accompanies both weak-doublet and weak-singlet realized matter sectors and survives any downstream admission of non-abelian internal rotation. In downstream recognition language this is the hypercharge sector, with overall generator normalization fixed at the structural level by the neutrality condition of the electroweak order parameter.

Definition 5 (Electroweak order parameter). The *electroweak order parameter* Φ_{EW} is the tachyonically forced vacuum/interface object whose nonzero realized value breaks the electroweak hosting fiber into charged, neutral-massive, and neutral-massless sectors.

Definition 6 (Minimal electroweak order doublet realization). At the minimal electroweak layer, the order parameter is realized in the minimal weak doublet host as

$$\Phi_{\text{EW}} = \begin{pmatrix} \phi_+ \\ \phi_0 \end{pmatrix},$$

with

$$T_3 \phi_+ = +\frac{1}{2} \phi_+, \quad T_3 \phi_0 = -\frac{1}{2} \phi_0.$$

The condensed component is ϕ_0 , and the surviving residual neutral charge must annihilate it:

$$Q_{\text{em}} \phi_0 = 0. \quad (1)$$

This is the electroweak-layer expression of the rule that a surviving zero-cost neutral direction is invisible to the realized condensate that selects the broken sector.

Transition Condition and Re-Articulation Rule

Definition 7 (Minimal realized electroweak layer). The *minimal realized electroweak layer* is the first matter/gauge hosting layer at which all of the following are simultaneously admitted: (i) local abelian phase redundancy from CF 00, (ii) a two-mode internal rotation sector from CF 09, (iii) an active chiral boundary-hosted matter family from CF 08/CF 13, and (iv) a tachyonically forced order parameter from CF 03.

Definition 8 (Electroweak-hosting fiber). The *electroweak-hosting fiber* is the minimal realized matter/gauge fiber that simultaneously hosts:

- (i) one abelian neutral companion direction, and
- (ii) one left-acting two-mode non-abelian internal rotation sector acting on a weak doublet-hosting pair.

Definition 9 (Neutral mixing basis and residual electromagnetic direction). The *neutral mixing basis* is the ordered pair of neutral gauge hosting directions before electroweak breaking,

$$(W^3, B),$$

where the first element is the neutral direction inside the minimal two-mode non-abelian weak sector and the second is the abelian neutral companion direction. The *residual electromagnetic direction* is the unique neutral massless zero-cost direction that remains after the electroweak order parameter is admitted and the neutral sector is diagonalized.

Definition 10 (Electroweak mixing angle). The *electroweak mixing angle* θ_{EW} is the rotation angle that diagonalizes the neutral broken-sector burden form in Definition 9 into the residual massless direction and the orthogonal massive neutral direction.

Definition 11 (Residual electromagnetic charge). Let T_3 denote the diagonal generator of the weak doublet-hosting pair and let Y denote the generator of the neutral companion burden. The *residual electromagnetic charge* is the unique unbroken neutral charge operator

$$Q_{\text{em}} := T_3 + \frac{Y}{2}, \quad (2)$$

obtained by fixing the companion-generator normalization so that the neutral component of the electroweak order parameter is uncharged under the residual massless direction.

Selection or Extremal Principle

Definition 12 (Selection principle). Realized electroweak evolution and sector decomposition follow the same stationary articulation-cost principle earned in CF 14: among admissible broken and unbroken realizations compatible with the inherited canon, the selected structure carries no avoidable first-order excess articulation burden.

This principle is not a new primitive. It is the same temporal effective invariant now applied to the gauge/matter layer that becomes available once the carrier, boundary-hosted chiral sector, multi-mode rotation sector, and order-parameter interface are simultaneously realized.

4 The Effective Stack or Layer Structure

The electroweak paper sits on a cumulative stack rather than a fresh beginning. Each layer below is already part of canon; CF 16 closes the first joint realization at which those inherited objects coexist in one minimal fiber.

Definition 13 (Effective electroweak stack). The ordered electroweak-relevant effective stack is

$$(A(-1), \text{CF000}, \text{CF00}, \text{CF03}, \text{CF08}, \text{CF09}, \text{CF13}, \text{CF14}, \text{CF15}, \text{CF16})$$

with each level carrying, respectively: primitive non-discharge; articulation/non-bearing logic; carrier and phase redundancy; tachyonic interface forcing; chiral boundary hosting; gauge hosting and multi-mode rotation; oriented chirality; stationary articulation-cost selection; zero-cost conservation and broken-sector logic; and the electroweak closure derived in the present paper.

Table 1: Electroweak stack and derivation pointers.

Level	Effective object	Structural expression	Source / paper
L0	Primitive invariant	non-discharge / re-articulation	A(-1), CF000
L1	Carrier and phase fiber	complex carrier + local phase	CF00
L2	Interface forcing	tachyonic vacuum relocation	CF03
L3	Chiral active matter	boundary-hosted oriented family	CF08, CF13
L4	Gauge hosting	abelian + multi-mode rotation	CF09
L5	Broken-sector logic	zero-cost / residual / normal mode	CF14, CF15
L6	Electroweak closure	$U(1)_{\text{comp}} \times SU(2)_L$ with mixing	CF16

The subtle point is L6: this is not a new primitive gauge theory. It is the first realized layer where the inherited abelian companion direction, the minimal two-mode non-abelian rotation sector, the chiral boundary family, and the broken interface order parameter are all present at once.

5 Main Governing Law: Electroweak Closure

Lemma 1 (Minimal two-mode non-abelian factor). *At the minimal realized electroweak layer, the smallest non-abelian norm-preserving rotation acting transitively on a weak doublet-hosting pair is $SU(2)$.*

Proof sketch. By CF 09, admitting more than one degenerate fiber mode opens a multi-mode internal rotation sector not exhausted by the abelian phase direction. At the minimal layer of Definition 7, the active family is a two-mode family by Definition 3. The norm-preserving special-unitary rotations of a minimal complex two-mode family form $SU(2)$. $\square$

Lemma 2 (Chiral hosting restriction). *The active non-abelian rotation sector may act only on the oriented boundary-hosted family and not symmetrically on both handed families.*

Proof sketch. By CF 08, the wall construction localizes one chiral family on the active boundary while the opposite-handed partner is asymptotically separated from that same hosting region. By CF 13, this asymmetry is real orientation, not labeling. So only the active boundary family inhabits the common hosting region required for a shared non-abelian action. $\square$

Lemma 3 (Unique residual neutral generator). *At the minimal electroweak layer, the unique residual neutral generator that annihilates the condensed order-parameter component is*

$$Q_{\text{em}} = T_3 + \frac{Y}{2}, \quad (3)$$

after normalization of the companion generator by the order-parameter doublet.

Proof sketch. Write the most general surviving neutral generator as

$$Q = T_3 + cY.$$

Let the order-parameter doublet carry companion-burden eigenvalue y_Φ : $Y\phi_+ = y_\Phi\phi_+$, $Y\phi_0 = y_\Phi\phi_0$. By Definition 6, the condensed component satisfies $T_3\phi_0 = -\frac{1}{2}\phi_0$ and must remain invisible to the surviving residual neutral direction, so

$$0 = Q\phi_0 = \left(-\frac{1}{2} + cy_\Phi\right)\phi_0.$$

Hence $c = (2y_\Phi)^{-1}$. Defining the normalized companion generator $\tilde{Y} := Y/y_\Phi$ gives

$$Q = T_3 + \frac{\tilde{Y}}{2}.$$

Renaming $\tilde{Y}$ as Y fixes the companion normalization and yields (3). The factor 1/2 is therefore forced by the minimal weak-doublet hosting structure together with the neutrality condition of the condensed component. $\square$

Lemma 4 (Single-condensate neutral burden law). *A single electroweak condensate induces a neutral broken-sector burden form that is positive semidefinite and rank one.*

Proof sketch. The condensate Φ_{EW} sees neutral gauge-hosting directions only through one neutral covector $n = (a, -b)$ on the basis (W^3, B) : one neutral combination is seen by the condensate and one orthogonal combination is not. Because there is one condensate and therefore one independent neutrality condition, the invisible neutral sector is one-dimensional. The quadratic burden carried by the seen direction is therefore

$$\nu^2(n \cdot x)^2 = \nu^2 x^\top (n n^\top) x,$$

with $x = (W^3, B)^\top$ and $\nu = |\Phi_{\text{EW}}|$. This matrix is symmetric, positive semidefinite, nonzero for nontrivial breaking, and rank one because it is the outer product of a single nonzero covector with itself. $\square$

Lemma 5 (Weak-host isotropy and charged-sector coefficient identity). *Before neutral mixing with the companion abelian axis, the weak directions (W^1, W^2, W^3) carry one common weak-host coefficient in the quadratic burden induced by the electroweak order parameter.*

Proof sketch. By Lemma 1, the weak sector is the minimal $SU(2)$ host acting on a doublet. Within that host the invariant has no preferred internal weak orientation prior to neutral mixing: all three weak directions lie on the same minimal internal rotation orbit. This is the same rotational indifference logic already earned at lower levels in the continuous ORS completion of CF 00: once a minimal rotation sector is admitted, no direction inside that undistinguished orbit is privileged before an external companion axis is mixed in. Therefore the quadratic burden contributed by Φ_{EW} to the pure weak directions carries one common coefficient a^2 . The companion axis contributes only through neutral mixing with W^3 . $\square$

Theorem 1 (Electroweak closure law). *For any admissible realization at the minimal electroweak layer of Definition 7, all of the following hold:*

1. **Fiber decomposition.** *The minimal electroweak-hosting fiber is*

$$\mathcal{F}_{\text{EW}} \cong U(1)_{\text{comp}} \times SU(2)_L. \quad (4)$$

2. **Chiral weak coupling.** *The $SU(2)$ factor acts only on the active chiral boundary sector and therefore becomes a left-acting weak sector.*

3. **Neutral companion survival and charge allocation.** *The inherited abelian phase burden of CF 00 persists as a unique neutral companion direction that partners the neutral weak direction in the broken neutral sector. Fixing the companion-generator normalization by requiring the neutral order-parameter component to be uncharged under the surviving massless direction yields the residual electromagnetic charge law*

$$Q_{\text{em}} = T_3 + \frac{Y}{2}. \quad (5)$$

4. **Neutral mixing law.** *For nonzero electroweak order amplitude $\nu := |\Phi_{\text{EW}}|$, the neutral broken-sector burden form in the basis (W^3, B) is rank-one positive semidefinite and takes the form*

$$\mathcal{M}_N^2 = \nu^2 \begin{pmatrix} a^2 & -ab \\ -ab & b^2 \end{pmatrix}, \quad (6)$$

with nonnegative structural coefficients a, b .

5. **Residual electromagnetic direction.** The neutral sector diagonalizes into one massless residual direction

$$A = \sin \theta_{\text{EW}} W^3 + \cos \theta_{\text{EW}} B, \quad (7)$$

and one orthogonal massive neutral direction

$$Z = \cos \theta_{\text{EW}} W^3 - \sin \theta_{\text{EW}} B, \quad (8)$$

where

$$\cos \theta_{\text{EW}} = \frac{a}{\sqrt{a^2 + b^2}}, \quad \sin \theta_{\text{EW}} = \frac{b}{\sqrt{a^2 + b^2}}. \quad (9)$$

6. **Charged/neutral mass law.** The charged weak directions acquire equal mass

$$m_W^2 = \nu^2 a^2, \quad (10)$$

the massive neutral direction has

$$m_Z^2 = \nu^2 (a^2 + b^2), \quad (11)$$

and therefore

$$m_W = m_Z \cos \theta_{\text{EW}}. \quad (12)$$

7. **Broken gauge–scalar split.** The only physical scalar remaining in the minimal realized layer is the normal mode of Φ_{EW} ; tangential broken directions are carried by the gauged sector and do not survive as additional physical scalars.

Contrapositive. Falsification of any listed clause falsifies the electroweak closure law.

Proof. Clauses (1) and (2) follow immediately from the two preparatory hosting lemmas. The abelian factor is forced because CF 00 already earns the local phase redundancy of the complex carrier; no later realized gauge-hosting sector may erase that previously earned neutral companion direction. By Lemma 1, the smallest non-abelian norm-preserving rotation on a minimal weak doublet-hosting pair is $SU(2)$. By Lemma 2, that action is restricted to the active boundary-hosted family, so the minimal hosting fiber is $U(1)_{\text{comp}} \times SU(2)_L$.

Clause (3) is fixed by the neutrality of the condensate, not by convention. The companion abelian direction persists because the non-abelian host refines the fiber structure rather than replacing the already-earned phase redundancy. At the minimal electroweak layer, write the most general surviving neutral generator as $Q = T_3 + cY$. By Definition 6, the condensed component is the lower component of the minimal weak doublet, so $T_3 \phi_0 = -\frac{1}{2} \phi_0$, and the surviving neutral zero-cost direction must annihilate it: $Q \phi_0 = 0$. By Lemma 3, this forces

$$Q_{\text{em}} = T_3 + \frac{Y}{2},$$

after normalization of the companion generator by the order-parameter doublet. So the residual electromagnetic charge law is earned inside the electroweak layer rather than imported from later notation.

Clause (4) is the A(-1) content of neutral electroweak breaking. Once $\Phi_{\text{EW}} \neq 0$, the neutral basis (W^3, B) is no longer the physical eigenbasis. Because one condensate constitutes one neutral saturation event, the condensate can see only one neutral combination while one orthogonal neutral combination remains invisible as the surviving residual zero-cost direction. By Lemma 4, the broken neutral quadratic burden form is necessarily the outer product of a single nonzero neutral covector with itself, so it is symmetric, positive semidefinite, nonzero, and rank one. Writing the seen neutral covector as $n = (a, -b)$ therefore gives

$$\mathcal{M}_N^2 = \nu^2 n n^\top = \nu^2 \begin{pmatrix} a^2 & -ab \\ -ab & b^2 \end{pmatrix},$$

which is exactly (6).

Clause (5) follows by diagonalizing the rank-one neutral form. The kernel of (6) is the span of (b, a) , so the unique surviving massless neutral direction is

$$A = \frac{b}{\sqrt{a^2 + b^2}} W^3 + \frac{a}{\sqrt{a^2 + b^2}} B,$$

which is (7) after identifying $\sin \theta_{\text{EW}} = b/\sqrt{a^2 + b^2}$ and $\cos \theta_{\text{EW}} = a/\sqrt{a^2 + b^2}$. The orthogonal normalized vector is $(a, -b)/\sqrt{a^2 + b^2}$, which gives the massive neutral direction (8). The mixing angle is therefore the actual diagonalizing angle of the one-broken/one-unbroken neutral decomposition, not an external parameter.

Clause (6) now follows from weak-host isotropy. Before mixing with the companion abelian axis, (W^1, W^2, W^3) lie on one undistinguished minimal weak orbit. By Lemma 5, the quadratic burden induced by Φ_{EW} assigns one common weak-host coefficient a^2 to the pure weak directions. The charged directions therefore acquire equal mass $m_W^2 = \nu^2 a^2$. The broken neutral direction carries both the weak and companion pieces of the seen neutral covector $(a, -b)$, so its mass is $m_Z^2 = \nu^2(a^2 + b^2)$. Combining these with (9) yields

$$m_W = \nu a = \nu \sqrt{a^2 + b^2} \frac{a}{\sqrt{a^2 + b^2}} = m_Z \cos \theta_{EW},$$

which is (12).

Clause (7) is the broken gauge–scalar split already closed abstractly in CF 15, now applied to the electroweak layer. By CF 03 the order parameter exists because the symmetric hosting configuration becomes tachyonically unstable and relocates burden onto an interface/vacuum manifold. By CF 15, fluctuations about a broken gauge-hosting sector split into tangential orbit directions and a normal direction. The tangential directions are carried by the gauged broken sector, while the normal direction changes the amplitude of Φ_{EW} away from the broken interface and remains as the only physical scalar. Therefore all seven clauses hold. $\square$

Corollary 1 (Standard electroweak recognition). *The abelian companion factor is recognized downstream as the hypercharge sector, the left-acting non-abelian factor as $SU(2)_L$, the massless neutral residual direction as electromagnetism, the massive neutral direction as the Z sector, the charged weak directions as $W^\pm$, and the normal interface mode as the Higgs scalar.*

Remark 1 (Why the neutral mixing angle is not imported rotational language). The diagonalizing angle θ_{EW} is a realized descendant of the same rotation logic already earned in CF 00. There the continuous ORS completion and the primitive half-turn constant π close angular structure as an earned consequence of orthogonal re-articulation rather than an imported geometry. At the electroweak layer the neutral pair (W^3, B) is simply another realized two-axis rotation problem: one direction is seen by the condensate and one orthogonal direction is not. The Weinberg angle is the neutral-sector rotation that expresses that broken/unbroken decomposition in the inherited basis.

Remark 2 (Scope lock: what CF16 closes and what CF18 may only refine). The Weinberg angle and the W/Z mass law are *closed in CF16*. They belong to the present paper because they are part of the minimal electroweak architecture proved in Theorem 1. CF18 does *not* own their existence, their structural definitions, or the tree-level relation (12); CF18 may only derive scale-dependent radiative corrections to quantities already closed here.

6 Forward Predictions and Program Map

The electroweak closure law implies the following forward predictions and maps them to later derivation targets.

Table 2: CF16 predictions and derivation pointers.

Prediction	Source / paper	Status
Minimal EW fiber	This paper	Closed at structural level
Chiral weak coupling	This paper	Closed at structural level
Residual electromagnetism	This paper	Closed at structural level
Tree-level mixing angle and W/Z law	This paper	Closed at structural level
Radiative corrections / running	CF18	Out of scope for CF16

P1 — Minimal electroweak hosting is fixed

Any realized layer satisfying Definition 7 must reduce to the fiber of Theorem 1. Any larger fiber is therefore a downstream extension rather than the minimal electroweak layer.

P2 — Weak coupling is necessarily chiral

Any realization that couples the weak non-abelian sector symmetrically to both handed boundary-hosted families violates the wall-localization and oriented chirality canon of CF 08/CF 13.

P3 — One neutral direction must remain massless

Any electroweak realization with no neutral zero-cost residual direction is incompatible with the rank-one broken neutral law in Theorem 1.

P4 — The charged/neutral mass relation is structural

Any realization of the present electroweak layer in which the same order parameter lifts the charged and neutral weak sectors but fails to obey (12) must violate one of the hypotheses of Theorem 1.

P5 — The physical scalar is unique in the minimal layer

Any additional electroweak physical scalar beyond the normal mode of Φ_{EW} requires either a higher-dimensional order parameter or additional sectors not included in the minimal realized layer of this paper.

7 Falsification Criteria

The electroweak closure law is falsified by any of the following:

- F1. Carrier/phase failure.** A realized electroweak layer satisfying Definition 7 does not admit an inherited abelian companion burden.
- F2. Chirality failure.** The active weak non-abelian sector can act on both handed boundary families without violating CF 08/CF 13.
- F3. Neutral residual failure.** The broken neutral burden form has no massless residual direction.
- F4. Mass-law failure.** The same electroweak order parameter lifts the charged and neutral weak sectors but violates (12).
- F5. Scalar-sector failure.** The only scalar remaining after minimal gauged electroweak breaking is not the normal mode of Φ_{EW} .
- F6. Recognition failure.** The standard electroweak recognition of Corollary 1 cannot be recovered as the coarse downstream reading of the VDM structure proved in this paper.

8 Validation Gates

Analytical Gates

Gate: G1: Minimal electroweak-hosting fiber *Threshold: Theorem 1, clauses 1–3 sound.*

The proof must show that the abelian companion factor and the left-acting minimal two-mode non-abelian factor are both forced and jointly minimal.

Status: Proved structurally in Theorem 1.

Gate: G2: Chiral weak coupling *Threshold: Theorem 1, clause 2 sound.*

The proof must derive left-only weak coupling from the domain-wall and chirality/orientation canon, not from imported representation theory.

Status: Proved structurally in Theorem 1.

Numerical or Programmatic Gates

Gate: G3: Neutral mixing gate *Threshold: Theorem 1, clauses 4–5 sound.*

The neutral sector must diagonalize into one exactly massless and one massive direction, with mixing angle defined by the diagonalizing rotation.

Status: Proved structurally in Theorem 1; no precision phenomenology claimed in this paper.

Gate: G4: Charged/neutral mass gate *Threshold: Equation (12) earned from the same order parameter.*

The charged and neutral massive sectors must be descendants of the same order parameter and satisfy the structural mass relation.

Status: Proved structurally in Theorem 1.

Empirical or Supporting Evidence Gates

Gate: G5: Residual electromagnetic gate *Threshold: Massless residual neutral direction survives.*

The surviving neutral massless direction must be the conserved residual abelian sector acting across the realized matter sectors.

Supporting evidence: Proved structurally in Theorem 1 and recognized in Corollary 1.

Gate: G6: Electroweak scalar gate *Threshold: Only the normal mode remains physical in the minimal layer.*

The physical scalar must be the normal mode of the broken electroweak interface; tangential directions may not remain as extra physical scalars within the minimal realized layer.

Supporting evidence: Proved structurally in Theorem 1.

Pass/Fail Criteria

- **PASS:** The electroweak hosting fiber, chiral weak coupling, neutral mixing, structural Weinberg angle, structural tree-level W/Z mass law, residual electromagnetism, and interface scalar closure are all derived from inherited canon without imported Standard Model primitives.
- **FAIL:** Any clause of Theorem 1 depends on an unearned primitive or collapses under the listed falsifiers.
- **Current status:** Closed at the declared CF16 scope: the structural/tree-level electroweak layer is derived and no internal electroweak burden is left open.

9 Discussion

Why This Result Sits at the Foundational Level

This paper sits at the first realized gauge/matter layer where the previously separate canon branches become one minimal hosting structure. It is foundational because it closes the first forced electroweak decomposition from A(-1)-descended objects already earned in the stack.

Interpretive Consequences

The result says the electroweak pattern is not a free representation choice. Once the inherited abelian phase direction, the minimal two-mode non-abelian rotation sector, the chiral boundary-hosted family, and the broken interface order parameter are admitted together, the qualitative electroweak architecture is forced. The standard labels are therefore recognition layers, not proof primitives.

Connection to Later Physics / Mathematics / Runtime Work

Later closures may sharpen flavor, confinement, and radiative corrections, but they do not replace the present structure. In particular, CF18 may only correct the already-closed CF16 tree-level quantities; it does not own the existence of the electroweak angle or the structural W/Z law. In runtime terms, this paper supplies a closed broken/unbroken sector architecture for any later VDM implementation that wants to express electroweak structure without importing it.

Programmatic Outlook

The next step is not to reopen CF16. The electroweak structural layer is closed here. Downstream work now splits cleanly: CF17 handles non-abelian confinement and the strong-force branch; CF18 handles

radiative/effective-action refinements, running couplings, and scale-dependent corrections to the present tree-level electroweak relations.

Open Questions

- **Within the declared scope of CF16, no open questions remain.**
- **Downstream refinement branch 1.** Strong-force quantitative closure belongs to CF17, not CF16.
- **Downstream refinement branch 2.** Radiative and running-coupling refinements belong to CF18, not CF16.

10 Conclusions

Starting only from already-earned CF canon, this paper derives the minimal electroweak-hosting fiber, the left-handed restriction of the weak non-abelian sector, the persistence of an abelian hypercharge companion burden, the electromagnetic charge law $Q_{\text{em}} = T_3 + Y/2$, neutral mixing into one massless and one massive direction, the structural/tree-level Weinberg angle, the structural/tree-level charged/neutral mass relation $m_W = m_Z \cos \theta_{\text{EW}}$, the residual electromagnetic direction, and the physical scalar as the normal mode of the broken electroweak interface.

Therefore CF16 is closed: the structural/tree-level electroweak layer — including the Weinberg angle and the tree-level W/Z mass relation — is derived from prior canon without imported Standard Model primitives. Later descendants may refine confinement or radiative corrections, but they do not reopen or repossess the closure established here.

References

- [1] J. K. Lietz. A(-1): Primitive Bifurcation Conjecture — Effective Origin, Non-Discharge, and Orthogonal Re-Articulation as Base Law of VDM. *Neuroca, Inc.*, 2026.
- [2] J. K. Lietz. CF000: Complete Formalism — Primitive Distinguishability and Orthogonal Articulation. *Neuroca, Inc.*, 2026. DOI: 10.5281/zenodo.19098778.
- [3] J. K. Lietz. CF00: Complete Formalism — Induced Geometry and Emergent Dynamics. *Neuroca, Inc.*, 2026.
- [4] J. K. Lietz. CF03: Complete Formalism — Universal Scaling, Tachyonic Interfaces, and Hierarchical Saturation in VDM. *Neuroca, Inc.*, 2026.
- [5] J. K. Lietz. CF08: Complete Formalism — Spinor Emergence via Domain-Wall Fermions in VDM. *Neuroca, Inc.*, 2026.
- [6] J. K. Lietz. CF09: Complete Formalism — Gauge Field Emergence via Berry Connection in VDM. *Neuroca, Inc.*, 2026.
- [7] J. K. Lietz. CF13: Complete Formalism — π , Transcendence, Chirality, and the Nielsen–Ninomiya Obstruction in VDM. *Neuroca, Inc.*, 2026. DOI: 10.5281/zenodo.19121368.
- [8] J. K. Lietz. CF14: Principle of Stationary Action as Temporal Effective Invariant. *Neuroca, Inc.*, 2026.
- [9] J. K. Lietz. CF15: Noether’s Theorem as Invariant Articulation Cost Under Symmetry. *Neuroca, Inc.*, 2026. DOI: 10.5281/zenodo.19153908.